

# Establishment of a Vegetation Cover on Tundra Kimberlite Mine Tailings: 2. A Field Study

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## Abstract

High erosion potential of dewatered kimberlite mine tailings after diamond extraction has prompted research at the Ekati Diamond Mine in the Canadian subarctic heath tundra ecosystem. Greenhouse and field studies aimed at establishing a permanent vegetation cover on these dewatered tailings began in spring 2000. Coarse texture, no organic component, lack of available macronutrients, and a serpentine chemistry are the principal limitations of kimberlite tailings to plant colonization. Structure-improving (peat moss, lake sediment, and sewage sludge) and nutrient-providing (fertilizer, rock phosphate, calcium carbonate, and gypsum) amendments were tested to ameliorate these conditions, facilitating the establishment of a permanent vegetation cover, which stabilizes surface

materials and promotes natural colonization by the surrounding tundra vegetation. Seven native grass species (*Arctagrostis latifolia*, *Calamagrostis canadensis*, *Poa glauca*, *Poa alpina*, *Deschampsia beringensis*, *Deschampsia caespitosa*, and *Festuca rubra*) were used to measure amendment success. With the addition of structure-improving and nutrient-providing amendments, plant growth on these kimberlite tailings under field conditions was significantly improved over unamended tailings material. Tailings properties, including cation exchange capacity, organic carbon, and macronutrient availability, were also improved with amendment addition.

**Key words:** amendments, diamond mine, native species, reclamation, revegetation, soil nutrients, soil structure.

## Introduction

Kimberlite tailings are the primary waste product of diamond mine operations in northern regions. Coarse texture (80% sand), no organic component, lack of available macronutrients, and a serpentine state are the principal limiting factors to plant growth on kimberlite tailings (Miles & Tainton 1979a, 1979b). Little is known of the ability of plant species native to subarctic tundra communities to establish and survive on such altered substrates. Greenhouse research has shown that through additions of nutrient-providing and structure-improving amendments, these limitations can be alleviated and substantial growth of native grasses is possible (Reid & Naeth 2005). The cool, dry climate and short growing season of the northern region, however, may impact the effect of amendments on kimberlite tailings and the performance of native plant species. A field experiment was established on kimberlite mine tailings at Ekati Diamond Mine, Northwest Territories (NWT). The study investigated the stabilization of the kimberlite tailings with a vegetation cover, focusing on kimberlite substrate improvement in relation to plant growth as well as native species tolerance of tailings conditions.

## Materials and Methods

### Site Description

Ekati Diamond Mine is located approximately 300 km north of Yellowknife, NWT, Canada. The mine is located in a heath tundra ecosystem characterized by boulder fields, sand eskers, and countless lakes. The area receives approximately 160 mm of precipitation annually, which almost exclusively falls between the months of July and October. Temperatures range between  $-40$  and  $25^{\circ}\text{C}$ , with only 4 months having mean temperatures over  $0^{\circ}\text{C}$  (BHP Diamonds, Inc. 1997).

Beginning October 1999, kimberlite tailings at Ekati Diamond Mine were pumped from the processing facility and deposited into the natural basin of Long Lake Containment Facility (LLCF), approximately 15 km away. By spring 2000, tailings at the most northerly end of LLCF had significantly consolidated. As a large exposed surface area without a protective surface cover, these dried kimberlite tailings had a high erosive potential, and stabilization became necessary.

### Experimental Design and Treatments

Field plots were established on the Ekati kimberlite tailings dewatered in LLCF in spring 2000. The study used a complete randomized block design replicated three times for both small plots and species trial study plots. Nine structure-improving and/or nutrient-providing amendment combinations were tested. Seven native grass species, supplied by Prairie Seeds, Inc., in Nisku, Alberta

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(Polar grass [*Arctagrostis latifolia* (R. Br.) Beauv.], Blue-joint [*Calamagrostis canadensis* Michx.], Tundra bluegrass [*Poa glauca* Beauv.], ARC Glacier alpine bluegrass [*Poa alpina* L.], Hulten Bering's tufted hairgrass [*Deschampsia beringensis*], Tufted hairgrass [*Deschampsia caespitosa* L.], and Common creeping red fescue [*Festuca rubra* L.]) were used to evaluate amendment success and their suitability for large-scale reclamation.

Two plot sizes were used: small plots of  $2 \times 2$  m to investigate amendment performance and species trial plots of  $2 \times 5$  m to investigate plant species tolerances to the amendments. In the first field season (August 2000), plots were established on dewatered kimberlite tailings at the north end of LLCF. Small plots and species trial plots were randomly assigned treatment applications. An unamended seeded treatment was included in the experimental design. Plots were surrounded by unamended and unseeded kimberlite tailings, against which amendment success was visually assessed.

Structure-improving amendments (peat moss, lake sediment, sewage sludge) were applied by volume. Peat was applied at 4 cm, lake sediment at 10 cm, and sewage sludge at 10 cm. Application rates were based on substrate limitations and maximum feasible quantities for large-scale field application. Structure-improving amendment materials were spread over the appropriate plots, hand-raked level, and incorporated into the tailings with a rototiller, creating a 28-cm-deep tilled surface cover. Treatment areas not receiving structure-improving amendments were also tilled to a 28-cm depth. Each plot was rototilled twice in opposite directions to ensure consistent mixing.

Nutrient-providing amendments were applied by weight and spread evenly over each plot with a drop spreader, then rototilled to a 17-cm depth into the plot surface. Inorganic fertilizer (by weight 10% N, 34% P, 10% K, and 5% S) was applied at 280 kg/ha, gypsum at 2,173 kg/ha, rock phosphate at 2,080 kg/ha, and calcium carbonate at 1,667 kg/ha. Once all the treatments were applied, the plots were seeded. The seed mix consisted of an equal number of seeds of each of seven native grass species. The mix was made by first establishing viability per 100 seeds for each species. Seeds were counted to establish a weight per 100 seeds of each species. The mix was then made on a per-species weight basis (different weights of each species), with an equal number of viable seed for each species.

Small plots were broadcast seeded at 40 kg/ha, combined with fertilizer. The small plots were lightly hand-raked to incorporate the seed and fertilizer into the surface, then covered with a Curlex (American Excelsior Company, Arlington, TX, U.S.A.) erosion blanket anchored to the kimberlite surface with 10-cm spikes to prevent surface erosion and amendment contamination among plots. In the species trial plots, individual species were hand-seeded at a 1- to 2-cm depth in randomly assigned rows. Three replicate rows of each species were sown for a total of 21 seed rows. After seeding, fertilizer

was broadcast over each species trial plot at 280 kg/ha. Application method and seeding rates were based on recommendations for cold soils (Kubanis 1982).

#### Soil Measurements

Soil was sampled randomly in the small plots at depth increments of 0–10, 10–20, 20–30, and 30–60 cm in both 2000 and 2001. Each sample was analyzed for available nutrients ( $\text{NO}_3$ , P, K, S), pH, total cations (Ca, Mg, Na, K), cation exchange capacity (CEC), extractable cations, total carbon, and total inorganic carbon. Available P and K were determined using a modified Kelowna extraction (Qian et al. 1994), available nitrate and S by calcium chloride extraction (Carter 1993), substrate pH by saturated soil paste (McKeague 1978), total cations by saturated paste (Carter 1993), CEC and extractable cations by ammonium acetate extraction (McKeague 1978), total carbon by Leco furnace combustion (Carter 1993), and total inorganic carbon by hydrochloric acid dilution (Carter 1993). Total organic carbon was calculated from the difference between total carbon and total inorganic carbon.

#### Vegetation Measurements

Treatment effectiveness was based on plant height, plant development (tillering and flowering), vegetation cover (ground and row in species trial plots), and shoot biomass. A  $1 \times 1$ -m area in the center of the small plots was randomly sampled three times using a  $20 \times 50$ -cm quadrat. In the species trial plots, random positions along each seeded row were selected for assessment with a  $20 \times 50$ -cm quadrat. The first assessment was completed at the end of August 2000 and the second in August 2001. Biomass was only measured in the second year due to the destructive sampling method.

Mean plant height was based on all individuals rooted in the quadrat, and maximum plant height was that of the tallest individual. Flowering and tillering were noted for individuals within each quadrat. Ground cover was estimated as a percentage of total area of each quadrat covered by vegetation. Biomass was measured, for both small and species trial plots, by randomly placing the  $20 \times 50$ -cm quadrat within each plot and clipping all vegetation rooted inside the quadrat at ground level. Shoot biomass was oven-dried at  $90^\circ\text{C}$  for 48 hours and weighed to determine dry biomass.

#### Statistical Analyses

Amendment performance within each plot type was compared using orthogonal contrasts (Petersen 1994), allowing the effect of individual amendments to be considered for each plant parameter measured. Duncan's multiple range test was used to make overall amendment comparisons. *T* tests were used to make more specific comparisons

between individual amendment combinations. A probability value of  $\leq 0.05$  was used.

## Results and Discussion

### Amendment Effect on Kimberlite Tailings Properties

Sewage sludge significantly increased available nitrate at 10- to 20- and 20- to 30-cm depths, resulting in highest available nitrate, 45 and 54 mg/kg, respectively (Table 1). Nitrate availability generally increased with depth in the tailings, and highest available nitrate was in sewage sludge, peat moss, gypsum, and rock phosphate treatments at 20–30 cm. P availability was significantly increased with sewage sludge ( $F = 0.0001$ ) at all depths. Peat moss significantly increased P availability at 0–10 cm ( $F = 0.0001$ ) and 20–30 cm ( $F = 0.0075$ ) but not at 10–20 cm ( $F = 0.2449$ ). Field availability of K and S was not significantly affected by any amendment.

CEC was significantly increased by peat moss at 10–20 cm ( $F = 0.0001$ ) and 20–30 cm ( $F = 0.0006$ ) but not at 0–10 cm ( $F = 0.2349$ ). CEC was significantly decreased by lake sediment at 10–20 cm ( $F = 0.0001$ ) and at 20–30 cm ( $F = 0.0070$ ). Sewage sludge increased CEC but only significantly at 20–30 cm ( $F = 0.0074$ ) (Table 2). Organic carbon was significantly increased by sewage sludge and peat moss at all depths. None of the Ca sources significantly affected CEC.

The fraction of extractable calcium to magnesium was not significantly improved by any amendment. Peat moss and gypsum in combination increased available calcium to 22.6 meq/100 g, the highest of any combination. However, this did not result in a significant improvement in the calcium–magnesium fraction, which was 1.09 (Ca = 22.6 meq/100 g, Mg = 20.7 meq/100 g).

### Amendment Effect on Plant Growth

In the first field season, ground cover (32%) significantly increased with sewage sludge ( $F = 0.0003$ ), applied in combination with peat moss, gypsum, and rock phosphate (Table 3). The highest ground cover (38%) was produced with lake sediment, gypsum, and rock phosphate (Table 3). Lake sediment alone improved ground cover (33%) ( $F = 0.0001$ ) (Table 3). Peat moss significantly improved ground cover ( $F = 0.0792$ ). Overall, vegetation on lake sediment significantly outperformed that on peat moss ( $F = 0.0001$ ) but was not significantly different from that on sewage sludge ( $F = 0.3947$ ). In the second field season, lake sediment was not as effective, and highest ground cover (70%) was produced with sewage sludge, peat moss, gypsum, and rock phosphate (Table 4). Sewage sludge and peat moss combined significantly increased ground cover ( $F = 0.0001$ ).

Sewage sludge, peat moss, and lake sediment in combination significantly increased plant height (Table 3).

**Table 1.** Mean available nitrate, phosphorus, potassium, and sulfur (mg/kg) of amended kimberlite tailings in the second growing season.

| Amendment | Depth (cm) | Nitrate     | Phosphorus   | Potassium | Sulfur      |
|-----------|------------|-------------|--------------|-----------|-------------|
| C         | 0–10       | 1.7 (0.1)   | 28.7 (4.6)   | 185 (18)  | 215 (52)    |
| C         | 10–20      | 2.3 (0.1)   | 9.0 (0.0)    | 212 (26)  | 513 (150)   |
| C         | 20–30      | 3.4 (0.2)   | 10.0 (0.9)   | 198 (10)  | 414 (41)    |
| G         | 0–10       | 1.1 (0.1)   | 28.0 (5.3)   | 171 (12)  | 322 (374)   |
| G         | 10–20      | 1.3 (0.2)   | 10.0 (0.9)   | 205 (10)  | 481 (287)   |
| G         | 20–30      | 1.8 (0.1)   | 7.7 (0.9)    | 185 (22)  | 647 (92)    |
| L         | 0–10       | 1.7 (0.3)   | 20.3 (3.8)   | 147 (8)   | 754 (247)   |
| L         | 10–20      | 0.8 (0.1)   | 12.0 (2.2)   | 145 (11)  | 775 (115)   |
| L         | 20–30      | 0.8 (0.1)   | 8.7 (1.2)    | 166 (15)  | 765 (153)   |
| LGR       | 0–10       | 0.9 (0.1)   | 19.0 (2.6)   | 144 (14)  | 827 (216)   |
| LGR       | 10–20      | 0.7 (0.0)   | 10.3 (1.0)   | 137 (18)  | 906 (193)   |
| LGR       | 20–30      | 0.9 (0.1)   | 8.0 (0.8)    | 149 (16)  | 712 (146)   |
| PG        | 0–10       | 0.9 (0.1)   | 27.0 (5.6)   | 192 (11)  | 694 (154)   |
| PG        | 10–20      | 1.0 (0.1)   | 9.7 (0.9)    | 219 (24)  | 1,129 (136) |
| PG        | 20–30      | 1.2 (0.2)   | 9.3 (0.9)    | 209 (25)  | 1,001 (190) |
| PGC       | 0–10       | 1.8 (0.2)   | 26.7 (2.3)   | 171 (10)  | 808 (277)   |
| PGC       | 10–20      | 1.3 (0.3)   | 11.3 (0.9)   | 186 (14)  | 899 (241)   |
| PGC       | 20–30      | 1.5 (0.2)   | 10.0 (1.1)   | 176 (13)  | 901 (105)   |
| PSRG      | 0–10       | 14.3 (3.3)  | 140.7 (14.8) | 193 (4)   | 904 (193)   |
| PSRG      | 10–20      | 44.9 (11.0) | 77.0 (11.2)  | 170 (28)  | 822 (305)   |
| PSRG      | 20–30      | 54.1 (23.0) | 38.7 (6.3)   | 195 (24)  | 849 (178)   |
| R         | 0–10       | 0.9 (0.1)   | 19.7 (4.8)   | 181 (29)  | 509 (238)   |
| R         | 10–20      | 0.9 (0.1)   | 8.3 (2.6)    | 157 (24)  | 361 (182)   |
| R         | 20–30      | 1.2 (0.2)   | 8.0 (2.4)    | 156 (18)  | 432 (176)   |

Numbers in parentheses represents the mean standard errors. C = calcium carbonate; G = gypsum; R = rock phosphate; P = peat moss; S = sewage sludge; L = lake sediment.

**Table 2.** Mean properties of amended kimberlite tailings in the second growing season.

| <i>Amendment</i> | <i>Depth (cm)</i> | <i>Available Calcium (meq/100 g)</i> | <i>Available Magnesium (meq/100 g)</i> | <i>CEC (meq/100 g)</i> | <i>Total Organic Carbon (%)</i> |
|------------------|-------------------|--------------------------------------|----------------------------------------|------------------------|---------------------------------|
| C                | 0–10              | 14.7 (1.3)                           | 13.8 (0.8)                             | 15.9 (1.1)             | 0.33 (0.1)                      |
| C                | 10–20             | 16.7 (1.6)                           | 17.6 (1.6)                             | 18.1 (1.8)             | 0.44 (0.1)                      |
| C                | 20–30             | 15.3 (1.2)                           | 17.2 (0.3)                             | 16.8 (2.0)             | 0.26 (0.1)                      |
| G                | 0–10              | 17.1 (2.2)                           | 13.5 (1.7)                             | 15.1 (1.4)             | 0.27 (0.1)                      |
| G                | 10–20             | 16.7 (1.8)                           | 16.9 (1.1)                             | 17.1 (2.1)             | 0.33 (0.1)                      |
| G                | 20–30             | 14.8 (1.3)                           | 17.0 (1.4)                             | 16.0 (1.8)             | 0.32 (0.0)                      |
| L                | 0–10              | 15.4 (1.4)                           | 13.2 (2.4)                             | 12.4 (0.9)             | 0.17 (0.0)                      |
| L                | 10–20             | 13.7 (1.2)                           | 14.4 (3.8)                             | 10.4 (0.8)             | 0.14 (0.0)                      |
| L                | 20–30             | 15.2 (1.5)                           | 17.4 (3.3)                             | 13.7 (0.7)             | 0.20 (0.0)                      |
| LGR              | 0–10              | 16.8 (2.8)                           | 12.7 (2.2)                             | 12.0 (0.8)             | 0.10 (0.0)                      |
| LGR              | 10–20             | 14.0 (2.6)                           | 12.6 (2.5)                             | 9.1 (0.5)              | 0.07 (0.0)                      |
| LGR              | 20–30             | 13.8 (1.3)                           | 15.0 (3.3)                             | 11.9 (1.1)             | 0.17 (0.1)                      |
| PG               | 0–10              | 18.7 (0.6)                           | 15.5 (0.8)                             | 17.1 (1.2)             | 0.73 (0.2)                      |
| PG               | 10–20             | 22.6 (0.5)                           | 20.7 (1.2)                             | 23.4 (2.4)             | 1.73 (0.3)                      |
| PG               | 20–30             | 19.3 (0.4)                           | 19.2 (0.6)                             | 20.6 (2.2)             | 1.23 (0.3)                      |
| PGC              | 0–10              | 18.3 (1.6)                           | 14.9 (3.8)                             | 16.6 (1.8)             | 0.83 (0.1)                      |
| PGC              | 10–20             | 20.2 (2.1)                           | 19.2 (4.2)                             | 21.9 (2.2)             | 1.70 (0.2)                      |
| PGC              | 20–30             | 17.9 (1.4)                           | 18.1 (3.4)                             | 18.4 (2.1)             | 1.10 (0.1)                      |
| PSRG             | 0–10              | 19.4 (0.7)                           | 15.9 (0.8)                             | 14.7 (1.6)             | 1.10 (0.1)                      |
| PSRG             | 10–20             | 18.4 (1.3)                           | 17.6 (3.2)                             | 18.4 (1.6)             | 1.83 (0.3)                      |
| PSRG             | 20–30             | 20.7 (0.8)                           | 20.1 (0.7)                             | 22.7 (2.2)             | 1.77 (0.3)                      |
| R                | 0–10              | 17.8 (2.5)                           | 14.8 (3.4)                             | 15.2 (3.8)             | 0.14 (0.1)                      |
| R                | 10–20             | 14.8 (1.6)                           | 14.6 (3.7)                             | 13.6 (3.7)             | 0.10 (0.0)                      |
| R                | 20–30             | 13.8 (1.3)                           | 15.2 (3.3)                             | 13.7 (2.6)             | 0.13 (0.0)                      |

Numbers in parentheses represent the mean standard errors. C = calcium carbonate; G = gypsum; R = rock phosphate; P = peat moss; S = sewage sludge; L = lake sediment.

Tallest plants (11.9 cm) were on sewage sludge, peat moss, gypsum, and rock phosphate. Plants were almost twice as tall as on the next most effective amendment combination of lake sediment, gypsum, and rock phosphate (6.7 cm). In the second field season, plants on plots with lake sediment did not perform as well.

Shoot biomass was only measured in the second growing season, and the sewage sludge, peat moss, gypsum, and rock phosphate treatment produced the highest mean shoot biomass (336.2 g/m<sup>2</sup>) (Table 4). The next most favorable amendment was lake sediment, gypsum, and

rock phosphate (129.4 g/m<sup>2</sup>). Sewage sludge, peat moss, or lake sediment significantly increased shoot biomass, with biomass on sewage sludge outperforming that on lake sediment and peat moss. No Ca source had any significant effect on shoot biomass in either year (Tables 3 & 4).

#### Native Species Performance

In the first growing season, very little growth occurred on the species study plots, with no significant differences among species or kimberlite amendments (Table 5).

**Table 3.** Mean plant growth on amended kimberlite tailings in the first growing season.

| <i>Amendment</i> | <i>Ground Cover (%)</i> | <i><math>\bar{X}</math> Height (cm)</i> | <i>Maximum Height (cm)</i> |
|------------------|-------------------------|-----------------------------------------|----------------------------|
| Control          | 4 (1)                   | 3.8 (0.3)                               | 4.7 (0.5)                  |
| C                | 4 (2)                   | 0.6 (0.1)                               | 2.1 (0.2)                  |
| G                | 4 (1)                   | 0.7 (0.1)                               | 2.5 (0.3)                  |
| L                | 33 (6)                  | 1.9 (0.3)                               | 4.8 (0.4)                  |
| LGR              | 38 (4)                  | 2.8 (0.3)                               | 6.7 (0.8)                  |
| PG               | 17 (2)                  | 1.8 (0.5)                               | 5.1 (0.8)                  |
| PGC              | 16 (2)                  | 1.7 (0.2)                               | 3.9 (0.3)                  |
| PSRG             | 32 (5)                  | 3.9 (0.4)                               | 11.9 (1.5)                 |
| R                | 6 (2)                   | 1.5 (0.5)                               | 3.5 (0.8)                  |

Numbers in parentheses represent the mean standard errors. C = calcium carbonate; G = gypsum; R = rock phosphate; P = peat moss; S = sewage sludge; L = lake sediment.

**Table 4.** Mean plant growth on amended kimberlite tailings in the second growing season.

| <i>Amendment</i> | <i>Ground Cover (%)</i> | <i><math>\bar{X}</math> Height (cm)</i> | <i>Maximum Height (cm)</i> | <i>Shoot Biomass (g/m<sup>2</sup>)</i> |
|------------------|-------------------------|-----------------------------------------|----------------------------|----------------------------------------|
| Control          | 4 (2)                   | 12.4 (2.2)                              | 20.3 (4.8)                 | 21.2 (14.4)                            |
| C                | 4 (1)                   | 9.6 (1.4)                               | 21.0 (3.3)                 | 26.8 (13.8)                            |
| G                | 2 (1)                   | 4.5 (1.6)                               | 8.7 (3.2)                  | 5.3 (2.7)                              |
| L                | 18 (2)                  | 22.7 (1.3)                              | 39.2 (1.5)                 | 112.8 (39.1)                           |
| LGR              | 22 (4)                  | 24.9 (1.5)                              | 43.3 (1.8)                 | 129.4 (10.7)                           |
| PG               | 8 (2)                   | 17.9 (1.0)                              | 34.4 (1.8)                 | 45.5 (27.2)                            |
| PGC              | 13 (3)                  | 16.9 (1.3)                              | 36.3 (3.0)                 | 84.3 (32.4)                            |
| PSRG             | 70 (12)                 | 42.6 (5.8)                              | 61.8 (7.4)                 | 336.2 (88.1)                           |
| R                | 3 (1)                   | 8.3 (2.1)                               | 16.1 (4.2)                 | 16.2 (7.5)                             |

Numbers in parentheses represent the mean standard errors. C = calcium carbonate; G = gypsum; R = rock phosphate; P = peat moss; S = sewage sludge; L = lake sediment.

**Table 5.** Mean performance of seeded native grasses in the first growing season.

| Species                         | Ground Cover (%) | Height (cm) |
|---------------------------------|------------------|-------------|
| <i>Arctagrostis latifolia</i>   | 0.2 (0.0)        | 1.0 (0.0)   |
| <i>Calamagrostis canadensis</i> | 0.5 (0.0)        | 1.0 (0.0)   |
| <i>Poa glauca</i>               | 0.2 (0.0)        | 1.0 (0.1)   |
| <i>Poa alpina</i>               | 0.7 (0.1)        | 2.0 (0.1)   |
| <i>Deschampsia beringensis</i>  | 1.1 (0.0)        | 3.5 (0.2)   |
| <i>Deschampsia caespitosa</i>   | 1.1 (0.1)        | 2.3 (0.1)   |
| <i>Festuca rubra</i>            | 1.5 (0.1)        | 4.6 (0.5)   |

Numbers in parentheses represent the mean standard errors.

*Deschampsia beringensis*, *D. caespitosa*, and *F. rubra* performed best in this first season, with *F. rubra* producing the highest ground cover (1.5%). In the second growing season, a significant difference was observed among kimberlite amendments. Peat moss significantly improved plant growth over amendment combinations without peat moss ( $F = 0.0001$ ) (Tables 6 & 7). Calcium had no significant effect on plant growth ( $F = 0.1765$ ). *Deschampsia beringensis* was most successful producing 53% cover and 213.6 g/m<sup>2</sup> of shoot biomass on peat moss-amended plots. *Festuca rubra* also performed well, with high ground cover (33%) and biomass (168.3 g/m<sup>2</sup>) on peat moss-amended plots (Table 6). Performance of *F. rubra* was not reduced as dramatically as that of the *D. beringensis* without peat moss (Tables 6 & 7). *Festuca rubra* lost 7% cover and 51.7 g/m<sup>2</sup> of biomass, whereas *D. beringensis* lost 35% cover and 125.8 g/m<sup>2</sup> biomass in the absence of peat moss.

#### Improved Structure Limitations and Nutrient Deficiencies

Field results supported greenhouse results (Reid & Naeth 2005), showing that low available macronutrients and poor substrate structure of the Ekati kimberlite tailings are critical limitations to plant establishment and development. As hypothesized, plant growth in the field was most successful when both structure limitations and nutrient deficiencies were improved. Sewage sludge and peat moss in combination provided significant improvements to the tailings as a growth medium, increasing nutrient availability and resulting in the most successful ground cover, plant

**Table 6.** Mean performance of seeded native grasses on amendment combinations that included peat moss in the second growing season.

| Species                         | Ground Cover (%) | Height (cm) | Shoot Biomass (g/m <sup>2</sup> ) |
|---------------------------------|------------------|-------------|-----------------------------------|
| <i>Arctagrostis latifolia</i>   | 0 (0)            | 0 (0)       | 0.0 (0)                           |
| <i>Calamagrostis canadensis</i> | 5 (2)            | 24 (5)      | 5.9 (2)                           |
| <i>Poa glauca</i>               | 2 (1)            | 5 (2)       | 1.2 (1)                           |
| <i>Poa alpina</i>               | 11 (2)           | 7 (1)       | 63.1 (11)                         |
| <i>Deschampsia beringensis</i>  | 53 (18)          | 44 (4)      | 213.6 (59)                        |
| <i>Deschampsia caespitosa</i>   | 16 (5)           | 13 (5)      | 84.4 (20)                         |
| <i>Festuca rubra</i>            | 33 (11)          | 13 (2)      | 168.3 (25)                        |

Numbers in parentheses represent the mean standard errors.

development, and biomass production. Nutrient additions without structural improvements showed poor plant response and resulted in available nutrients concentrating deep in the tailings profile (30–60 cm). Without organic carbon, nutrients may be leached deeper within the profile and washed out of the rooting zone. Lake sediment without nutrient provision resulted in good initial plant response, but by the second year, growth had significantly decreased, further supporting the hypothesis that both structure improvement and nutrient provision will be critical in establishing a sustainable vegetation cover on these kimberlite tailings. Without a capital of nutrients, cycling cannot occur; therefore, initial nutrient additions will be essential for system self-sustainability.

Ca sources had no significant effect on plant response or substrate properties over two growing seasons. Research on plant response to gypsum additions on sodic mine spoil suggests that growth (cover and biomass) is not significantly affected until the third growing season, illustrating the time required for the gypsum to dissolve and become an effective soil nutrient (Schuman et al. 1994). In this study, the site received an average annual precipitation of 362 mm compared to the annual precipitation at the Ekati mine site of below 160 mm. It is likely that low precipitation combined with latitude and short growing season (BHP Diamonds, Inc. 1997) will further extend the time required for the calcium sources to show significant results.

The availability of N and P in the rooting zone is considered the most common limiting factor to plant growth on coarse-textured substrates. This research concurs with the most successful field amendment combination of sewage sludge, peat moss, gypsum, and rock phosphate providing available nitrate 20 times that of fertilized treatments, as well as sufficient organic carbon to hold these nutrients and facilitate their uptake. This sewage sludge and peat moss combination maintained available nitrate levels through the second growing season at a 10–20-cm depth.

#### Overall Amendment Success

In this research, sewage sludge and peat moss in combination outperformed all other amendment combinations in

**Table 7.** Mean performance of seeded native grasses on amendment combinations that did not include peat moss in the second growing season.

| Species                         | Ground Cover (%) | Height (cm) | Shoot Biomass (g/m <sup>2</sup> ) |
|---------------------------------|------------------|-------------|-----------------------------------|
| <i>Arctagrostis latifolia</i>   | 0 (0)            | 0 (0)       | 0 (0)                             |
| <i>Calamagrostis canadensis</i> | 4 (1)            | 18 (4)      | 3.2 (0)                           |
| <i>Poa glauca</i>               | 0 (0)            | 0 (0)       | 0 (0)                             |
| <i>Poa alpina</i>               | 3 (2)            | 7 (1)       | 12.3 (2)                          |
| <i>Deschampsia beringensis</i>  | 18 (8)           | 6 (2)       | 87.8 (26)                         |
| <i>Deschampsia caespitosa</i>   | 8 (4)            | 16 (6)      | 25.6 (13)                         |
| <i>Festuca rubra</i>            | 26 (9)           | 9 (4)       | 117.6 (22)                        |

Numbers in parentheses represent the mean standard errors.

almost all quantified parameters and, considering feasibility, is recommended for testing on a large-scale. Lake sediment performed well, but a reduction in positive vegetation response in the second season as well as poor performance in the greenhouse study (Reid & Naeth 2005) suggests that further exploration of the effects of lake sediment is required. The fine-textured lake sediment was hypothesized to improve the tailings structure, increasing water- and nutrient-holding capacities. However, this was not observed in either the field or greenhouse studies. In fact, lake sediment with increased application rate reduced CEC in greenhouse trials. The results of the field research suggest that lake sediment will not sustain acceptable plant growth over time.

The beneficial effects of sewage sludge on plant growth, however, may be overshadowed by the associated risks of using biosolid waste materials as a soil amendment (Dunigan & Dick 1980; Bresson et al. 2001). The high fecal coliform ( $6.8 \times 10^6$  CFU/dL) and *Streptococcus* ( $3.4 \times 10^6$  CFU/dL) levels in the Ekati sewage sludge may affect its use as a large-scale reclamation amendment. However, research on coliform survival in surface-applied sewage sludge suggests that the health threats posed by coliform levels are temporary because survival is quickly reduced with time (Edmonds 1976). Edmonds showed that with winter application, the rate of survival is rapidly reduced and after only 162 days, 0.02% of the original coliform levels are present.

In considering the long-term effects of adding amendments to the Ekati kimberlite tailings, a closer evaluation of C and N must be made. If a self-sustaining vegetation cover is to be established, conditions for the mineralization of N must be considered (Palmer 1987). Results from Reeder and Berg (1977) illustrate how substrate availability of ammonium and pH fluctuation affect microbial activity and consequently the availability of N to vegetation. The pH of the Ekati kimberlite tailings is 8 and should not be a factor in nitrification. However, the C:N ratio of amendments is an important factor.

Peat moss and papermill sludge were added as organic amendments and both had C:N ratios of greater than 25:1 (Reid & Naeth 2005). The addition of these materials will initially limit N; however, with time, N will be mineralized as C breaks down improving the C:N ratio for microbial activity. These organic amendments performed best when combined with a nutrient source, making the overall C:N ratio more favorable. The addition of organic matter to improve substrate structure must be matched with appropriate levels of N if amendments are to be efficiently utilized and long-term vegetation success is to be achieved.

#### Plant Species Success

In considering plant species as potential candidates for large-scale reclamation, both *F. rubra* and *D. beringensis* performed well. Both these species were selected based on their native origins, their reputation for tolerance to

low-fertility regimes, and their associated ability to survive on highly disturbed sites at northern latitudes (Mitchell 1986). *Festuca rubra*, a common arctic wetland species, is found throughout the circumpolar arctic region (Kidd 1999). At more southern latitudes, this species can present an invasive threat; however, at its natural latitude, it is considered a common and important component of many riparian plant systems (Kidd 1999). *Deschampsia beringensis* is also known for its persistence on disturbed sites and is commonly referred to as a good, low-maintenance ground cover species (Mitchell 1986).

All seeded native grasses, with the exception of *Arctagrostis latifolia*, which did not grow on the Ekati tailings, are recommended for use in a reclamation seed mix. Studies on succession in tundra reclamation suggest that the higher the diversity of natives used in reclamation seed mixes, the higher the incidence of colonization from surrounding undisturbed areas, resulting in a more natural successional path for reclaimed sites (Cargill & Chapin 1987). Therefore, in the interest of promoting colonization of native species from the surrounding undisturbed tundra, diverse seed mixes are recommended even if some species are present in low numbers. Continued investigation of the potential of other native grasses, forbs, and shrubs to establish on amended kimberlite tailings is desirable.

Monitoring of the field research plots should continue, considering that the effects of Ca have not yet been observed and self-sustainability cannot be sufficiently studied in two growing seasons. Amendment additions to these dewatered kimberlite tailings have yielded encouraging results, and when both structural and nutrient limitations are dealt with, plant growth sufficient to stabilize this substrate is feasible.

#### Conclusions

- (1) Plant growth under field conditions was most favorable when both structure and nutrient limitations of the Ekati kimberlite tailings were simultaneously addressed.
- (2) Peat moss performed well, increasing water- and nutrient-holding capacities and overall tailings structure.
- (3) Sewage sludge was the most effective amendment in nutrient provision and combined with peat moss produced the most favorable plant response.
- (4) Lake sediment showed promising initial plant growth but in only two seasons started to show a negative tailings property response, and subsequent reduced plant growth. Lake sediment should be further tested at reduced rates (<5 cm) in combination with an organic carbon source (peat moss or papermill sludge) and a nutrient source (sewage sludge).
- (5) Calcium additions did not significantly increase the amount of available calcium in ameliorating the tailings serpentine state. However, continued monitoring of field plots could reveal long-term benefits.

- (6) *Festuca rubra* and *Deschampsia beringensis* produced the best plant growth response. However, all species except *Arctagrostis latifolia* performed well and are recommended for use in the Ekati native reclamation seed mix.

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